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TEST REPORT

ULTRASONIC ECHO SOUNDER TEST ON VERTICAL SHAFT EXCAVATION (KODEN)

Test Standard: HKCI TM3 Issue 2 (2024) | In-house Method: STP02-FDT-TM3 rev 7

Report No.		**Job Ref. No.**	
Client		**Date of Issue**	
Project			
Site Location		**Contract / W.O. No.**	
Pile No. & Pile Type		**Date Tested**	
Test Procedure	STP02-FDT-TM3 rev 7 (Eff. 02 Feb 2026)	**Ambient / Fluid Temp. (°C)**	

1. Calibrated Equipment Traceability & Shaft / Machine Settings

Instrument	Equipment ID	Cal. Due Date	Shaft / UES Parameter	Recorded Value
Koden Echo Sounder	LT-KM-04		**Nominal Pile / Casing Dia. (NS / WE)**	_____ mm / _____ mm
100m Measuring Tape	LT-MT-06		**Platform / Cut-off / Toe Level (mPD)**	_____ / _____ / _____
Digital Caliper	LT-DC-01 / 03		**Winch Speed (5–11 m/min) & Feed Rate**	_____ m/min | 1 : _____
Dipmeter & Spirit Level	LT-DM-01 / LV-01		**Fluid Medium & Specific Gravity**	☐ Water ☐ Bentonite

2. Pile Depth Verification & Horizontal Scaling Factor S_H (0.0001 m/mm)

Depth by Weighted Tape (m)		**Depth on UES Profile (m)**		**Diff. T ≤ min(0.3m, 1%)**	☐ Pass ☐ Fail
Calibration Check	**Axis X-X' S_{Hx} (m/mm)**	**Axis Y-Y' S_{Hy} (m/mm)**	**Measured Shaft Dia. (m)**	**Rock Socket Dia. (m)**	**Bell-out Dia. (m)**
Mean of 3 Readings (ROUND 4 d.p.)					

3. Overall & 10 m Sectional Shaft Offset and Verticality Summary ($L = \Delta r \times S_H$)

Section	Level D_1–D_2 (mPD)	ΔD (m)	X-X' Offset L_x (mm)	Y-Y' Offset L_y (mm)	Resultant L_R (mm)	Direction	Verticality 1 : N
Overall Shaft D1–D2							
1st Sec: 0–10 m							
2nd Sec: 10–20 m							
3rd Sec: 20–30 m							
4th Sec: 30–40 m							
5th Sec: 40–50 m							
6th Sec: 50–60 m							

Measurement Uncertainty & Remarks: Evaluated per FC03-TM3-MU (ISO/IEC Guide 98-3, k = 2 at ~95% confidence): Expanded uncertainty on shaft dimension / offset ± 0.001 m (1 mm) on calibrated scaling.

Tested & Recorded By (Qualified Operator):

Name & Signature Date: _____

Checked & Approved By (HOKLAS Approved Signatory):

Authorized Panel: Lam Hoi Fan, James (JL) / Chu Lap Man, Raymond (RC) / Tang Man Fu (MFT)

Authorized Signature & Company Chop Date: _____

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